

Shared-Latent Multi-Fidelity ROM for CFD-to-Experiment Transfer

Methodology Draft

Project methodology presentation

May 8, 2026

Problem Setting

Final application

- Low fidelity: CFD aerosol trajectories
- High fidelity: experiments
- Both are high-dimensional and time-dependent
- A paired LF-HF subset is assumed available

Main difficulty

- CFD and experiments are not separated only by noise or resolution
- We expect nonlinear observation bias and mild evolution mismatch
- HF data are scarce, so the surrogate must use LF efficiently

Test Problem and Different-Grid Observation Model

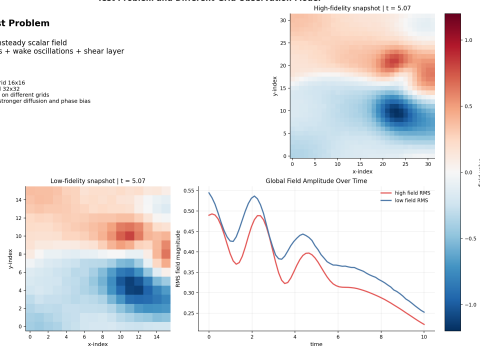
CFD-Like Test Problem

Parameterized unsteady scalar field
advected vortices + wake oscillations + shear layer

Observation model

- low fidelity: coarse grid 16x16
- high fidelity: fine grid 32x32
- same scalar quantity on different grids
- low fidelity includes stronger diffusion and phase bias

Dataset composition
paired = 4
low only = 1
high only = 0
 $\mu = (0.868, 0.854)$



Why Not Only a Two-Step Model?

Classical two-step baseline

$$(\mu, \eta) \mapsto \hat{x}^{\text{LF}}(t)$$
$$\hat{x}^{\text{HF}}(t) = G(\hat{x}^{\text{LF}}(t), \mu, \eta)$$

- LF is mainly an intermediate predictor
- HF stage learns a correction map
- Simple and strong as a baseline

Proposed shared-latent view

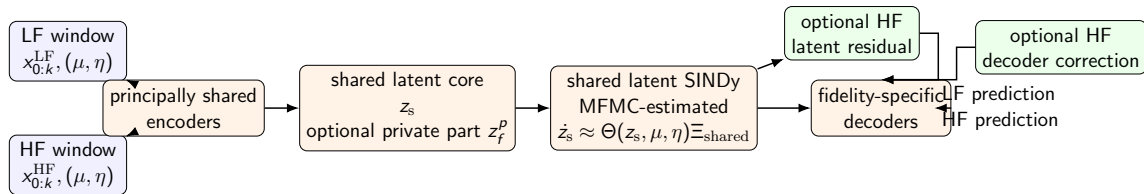
- LF should help learn the reduced dynamical backbone itself
- HF should calibrate what CFD misses
- Shared latent dynamics are the main scientific object

LF/HF $\rightarrow$ shared latent core $\rightarrow$ latent dynamics

Key difference

Two-step correction uses LF as an input to a transfer map. The shared-latent strategy uses LF to reduce the sample complexity of the latent dynamical model.

Scientific Backbone



- Shared core captures common aerosol dynamics
- Optional private/correction terms absorb CFD-to-experiment discrepancy
- Parameter and initial-condition information enter both encoding and latent dynamics

Latent Representation

$$E_f(x_{0:k}^{(f)}, \mu, \eta) \mapsto (z_s, z_f^p)$$

$$\hat{x}^{(f)} = D_f(z_s, z_f^p, \mu, \eta)$$

Interpretation

- z_s : shared latent dynamical core
- z_f^p : optional fidelity-specific residual information
- μ, η : operating parameters and initial-condition descriptors

Design choices

- Fully shared latent state
- Principally shared AE
- Shared core + private latent block
- Parameter/IC-informed encoding

Recommended default

Start with a shared core and keep the private block optional. This is flexible enough for CFD/experiment mismatch without giving up a common latent basis.

Paired LF-HF Data and Alignment

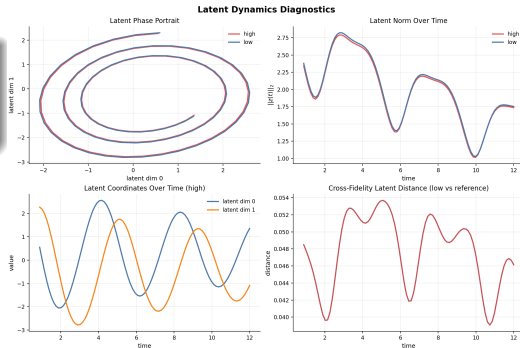
Because pairs are assumed available

We can align the shared latent core directly on matched operating conditions and matched times.

$$\mathcal{L}_{\text{align}} = \|z_s^{\text{LF}} - z_s^{\text{HF}}\|^2$$

$$\mathcal{L}_{\text{roll}} = \sum_t \|z_s^{\text{LF}}(t) - z_s^{\text{HF}}(t)\|^2$$

- Alignment is applied to the shared core, not necessarily to the whole latent code
- The penalty should remain soft because discrepancy is real



MFMC-SINDy on the Shared Latent Core

Shared latent dynamics model

$$\dot{z}_s \approx \Theta(z_s, \mu, \eta) \Xi_{\text{shared}}$$

HF target moments

$$C_{XX}^{\text{HF}} = \mathbb{E}[\Theta^\top \Theta], \quad C_{XY}^{\text{HF}} = \mathbb{E}[\Theta^\top \dot{z}_s]$$

MFMC estimators

$$\hat{C}_{XY}^{\text{MF}} = \hat{C}_{XY}^{\text{LF}} + A_{XY} \left(\hat{C}_{XY}^{\text{HF}} - \hat{C}_{XY}^{\text{LF},p} \right)$$

$$\hat{C}_{XX}^{\text{MF}} = \hat{C}_{XX}^{\text{LF}} + A_{XX} \left(\hat{C}_{XX}^{\text{HF}} - \hat{C}_{XX}^{\text{LF},p} \right)$$

Why this is natural

- SINDy is nonlinear in z_s but linear in Ξ_{shared}
- LF gives low-variance moment estimates
- Paired HF corrects LF bias
- HF data are spent on bias correction, not on learning everything from scratch

$$\Xi_{\text{shared}} = \arg \min_{\Xi} \|\hat{C}_{XX}^{\text{MF}} \Xi - \hat{C}_{XY}^{\text{MF}}\|_2^2 + \beta \|\Xi\|_2^2$$

Optional HF Corrections

If mismatch is mostly observational

$$\hat{x}^{\text{HF}} = D_{\text{HF}}(z_s, z_{\text{HF}}^p, \mu, \eta) + r_{\text{HF}}(z_s, \mu, \eta)$$

If mismatch enters the evolution

$$\dot{z}_s^{\text{HF}} \approx \Theta(z_s, \mu, \eta) \Xi_{\text{shared}} + \delta_{\text{HF}}(z_s, \mu, \eta)$$

Ablation order

- 1 Shared core only
- 2 Shared core + HF decoder correction
- 3 Shared core + HF latent residual
- 4 Neural ODE refinement only if needed

Reason

The first paper should isolate how far the shared MFMC-SINDy backbone already goes before adding a stronger HF-only corrector.

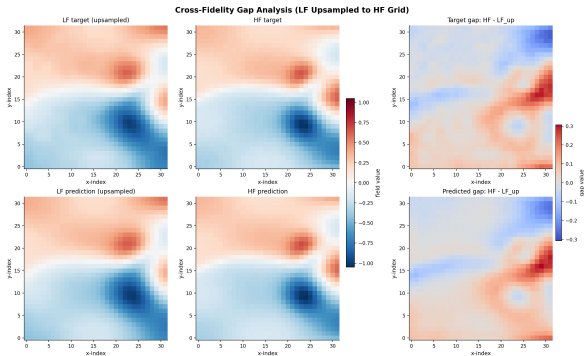
Staged Training Pipeline

- 1 Train the principally shared autoencoder on LF/HF trajectories with parameter conditioning.
- 2 Use paired data to align the shared latent core and compute latent rollout mismatch.
- 3 Encode trajectories and estimate latent derivatives.
- 4 Build the SINDy library on the shared core.
- 5 Fit the shared operator with MFMC-style moment estimators.
- 6 Fit an optional HF correction only if the shared core is insufficient.
- 7 Run ablations and compare against the two-step baseline.

Why staged training

- It matches the mathematical role of MFMC in latent linear regression
- It is more stable than full Neural ODE training from the start
- It keeps the scientific decomposition interpretable

Benchmark Illustrations



What the plots should show

- The LF-HF gap is structured, not random noise
- A useful model must capture both common evolution and fidelity discrepancy
- Evaluation must include cross-fidelity gap metrics, not only per-fidelity reconstruction

Core metrics

- reconstruction error
- paired latent alignment error
- latent SINDy residual
- rollout error

Comparison with the Two-Step Baseline

Aspect	Two-step LF→HF correction	Shared-latent MFMC-SINDy
Role of LF	Intermediate prediction to be corrected	Data source for latent manifold and latent dynamics
Use of paired data	Mostly for learning correction map	For latent alignment and MFMC bias correction
Common physics HF scarcity benefit	Implicit Mostly on correction stage	Explicit shared latent backbone On shared dynamical operator estimation
Interpretability	Limited	Shared core plus explicit residual decomposition
Training complexity	Lower	Higher, but more structured

Required baseline

The two-step model remains mandatory in experiments because it is simpler, strong, and may win if CFD and experiments are too different dynamically.

Development Priorities

Immediate engineering tasks

- define the paired-data contract and conditioning variables
- choose shared core only vs shared core + private latent block
- implement latent derivative estimation and SINDy library construction
- implement MFMC moment estimators and sparse latent solve
- add a minimal shared-latent example and a minimal two-step baseline

Validation tasks

- dataset and pairing tests
- SINDy library and derivative tests
- MFMC regression unit tests
- end-to-end pipeline smoke test
- example-level regression checks

Takeaway

- The chosen methodology is a paired shared-latent multi-fidelity ROM with MFMC-SINDy at the latent level.
- LF CFD trajectories are used to learn the dominant latent backbone, not only a pre-correction surrogate.
- HF experiments calibrate what CFD misses through paired alignment and optional HF residual correction.
- Neural ODE refinement remains possible, but it is not the first model to build.

Shared latent backbone first, HF correction second, Neural ODE later if needed.